

Compositional monitoring of energy-mix drift on the simplex

A compositional change-detection protocol read natively in Aitchison geometry

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AI Use Declaration — final slide · HUF-STD-001 compliant

The instrument reads. The expert decides. The hashes carry the receipts. The vocabulary holds the line.

Beat 1 — The simplex view

"Energy generation is a composition."

A country's electricity mix is a vector of carrier shares that sums to one.

Carriers (EMBER · 9 on the 8-simplex): coal · gas · oil · nuclear · hydro · wind · solar · bioenergy · other

Window 2000–2025 · Data: EMBER (CC BY 4.0)

Closure, perturbation, Aitchison distance, the simplex —
the geometry on which this entire talk operates.

Beat 2 — The data and the central claim

"No monitoring framework combines all three conjuncts."

Monitoring frame: Magnitude · Identity · Trend · Composition — this talk addresses Composition.

Aitchison-native

Read in the simplex's own geometry — closure, CLR, ILR-Helmert. Not approximated; not lifted into a Euclidean ambient.

Formal change detection

Metric / pivot displacement quantified with TV distance and K-eff. Trajectory captured, not just snapshot.

Carrier-level attribution

The helmsman channel names which carrier moved. Structural change is decomposable.

Combined into one observable stack.

This conjunction is the falsifiable claim of the talk.

Beat 3 — The protocol

"Perturbation, Aitchison distance, TV distance, K-eff."

Perturbation

compositional shift

classical CoDa operator

Aitchison distance

the simplex metric

classical CoDa operator

TV distance

structural velocity

small supporting metric

K-eff

concentration measure

small supporting metric

INV-050 CANONICAL · TV distance / Aitchison distance pair-invariance verified across 101 datasets.

Self-discipline note: L2 → TV correction is on the record. We tell on ourselves when we mislabel.

The mathematics is not new; the monitoring application may be.

Beat 3 / 10 · 2 min · conservative, well-documented

Beat 4 — Japan

"Japan, Fukushima, 2011–2012, the spike."

Step-Δ Aitchison

× 3

over neighbouring years

Trajectory: abrupt external shock

Carrier-level attribution: nuclear → fossil substitution post-2011.

Helmsman flips: 17 (highest in the 8-country corpus).

Deceptive-drift transitions: 2.

Protocol verdict: regime change. Shock loud and unambiguous.

Beat 5 — Germany • the headline

"Germany, the trajectory, the deceptive drift, $p = 0.0016$ — with the null caveat."

p-value

0.0016

Trajectory: continuous arc toward the renewable vertex.

IR class: OVERDAMPED_EXTREME (smooth convergence).

Helmsman flips: 13.

Reading at monthly grain • the packet's headline result.

Null-model caveat — read out loud:

Computed against the series' own empirical-frequency baseline.

This is a weaker null than a Dirichlet, permutation, or bootstrap null.

Beat 6 — United Kingdom

"The UK coal exit registers as a regime change."

Japan

Abrupt · external

shock

UK

Abrupt · policy-driven

regime change

Helmsman flips: 15

Germany

Continuous · policy-driven

drift

The protocol discriminates among three archetypes of structural change.

Beat 7 — Beyond the three

"5 of 9 countries reproduce the deceptive-drift signature."

Signature fires (5)

AUS · Australia
CHN · China
GBR · United Kingdom
IND · India
JPN · Japan

Does not fire (4)

DEU · loud drift at annual grain
FRA · loosening regime
USA · stable composition
WLD · aggregate signal

INV-051 CANONICAL · the protocol discriminates · it does not over-fire.

Beat 8 — Three open questions for the room

"I do not have these answers. I am asking you."

Q1

Concentration vs Aitchison norm

Are K-eff (concentration) and Aitchison-norm tightening the same observable, or different facets?

Q2

Right family of simplex distances

Which family of valid simplex distances is appropriate for testing verdict-invariance across the trajectory? (INV-050 follow-up)

Q3

Right null model for compositional change-point

What null is appropriate for the deceptive-drift p-value? Dirichlet? Permutation? Bootstrap? (INV-051 follow-up — methodologically consequential)

Beat 9 — How to defeat this claim

"A defeater must combine all three conjuncts."

Metric defeat

show the verdict flips under a different valid simplex metric
· preempted by INV-050

Case defeat

show signature fires uniformly everywhere (no discrimination) · preempted by INV-051

Prior-art defeat

show the three-conjunct conjunction already exists · Morais et al. 2017/18 · Arata & Onozaki 2017 closest adjacent · 3 areas pending · open

Category defeat

show this is an application note within existing CoDa, not a new monitoring category · we have no preconceived answer · open

A defeater must combine all three conjuncts. Inspection is invited.

Beat 10 — Close

"Two repositories, one self-discipline note, thank you."

higgins-decomposition (Hs)

Deterministic engines · CNT v3.1.0 / CNQ v2.0.0

63 Investigation Catalog entries · Hs Change Control v1.0

Higgins-Unity-Framework (HUF)

MC-4 framing · EITT · HUF Governance Charter

KILL-001 (19 named failure modes) · case studies

Self-discipline note: L2 → TV correction is on the record.

Hashes carry the receipts. Anyone with the raw CSV can verify any plate in ~2 minutes.

The talk is an ascent waypoint, not the summit.

Thank you. I'll take questions.

Q&A bench · extended material on disk

"What sits behind the slides if the room asks."

EITT

Entropy-Invariant Time Transformer — Shannon entropy conserved under geometric-mean temporal compression;
0.18% / 341:1. papers/EITT_CANONICAL_EXPLANATION_2026-05-12.md

Twin-quaternion factoring

INV-029 CANONICAL · D=8 factoring verified at IEEE machine floor (3.33e-16) on EMBER China.

CHSH joint-coherence

INV-035 CANONICAL · 0.88 within Tsirelson bound.

KILL-001

19 named failure modes · published falsifiability artifact (HUF, 2026-03-23).

HUF Governance Charter

9 articles (April 2026) · parent doctrine for Hs Change Control v1.0.

Bread the Hs way

verbal pedagogy · the dinner-table version of the framework · papers/BREAD_THE_HS_WAY_2026-05-12.md

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AI Use Declaration

per HUF Publication Standards (HUF-STD-001) — conforming to scientific community conventions

ICMJE · COPE · Nature/Springer · Science/AAAS · WAME · EU AI Act (2024) · arXiv · ACM · IEEE

AI tools used

Claude (Anthropic) · ChatGPT (OpenAI) · Copilot (Microsoft) · Gemini (Google) · Grok (xAI) — the HUF AI Collective

Tasks performed by AI

Drafting of supporting documents and slides · consistency editing · cross-checking claims across documents · literature-search assistance for prior-art areas · falsifiability-condition enumeration (KILL-001) · adversarial review for overclaim and drift catching

Author responsibility

The author (P. Higgins, Rogue Wave Audio) retains full responsibility for all scientific claims, data interpretation, methodological choices, and conclusions. All AI-generated content has been reviewed and verified. AI tools are NOT listed as authors.

AI use governance

HUF AI Collective cross-check protocol · HUF Governance Charter Articles II–IV · SAFE-001 cognitive-agent safety doctrine · dates of use: March – May 2026